Grade 9 Mathematics – Graphs

Answer Sheet – Worked Solutions

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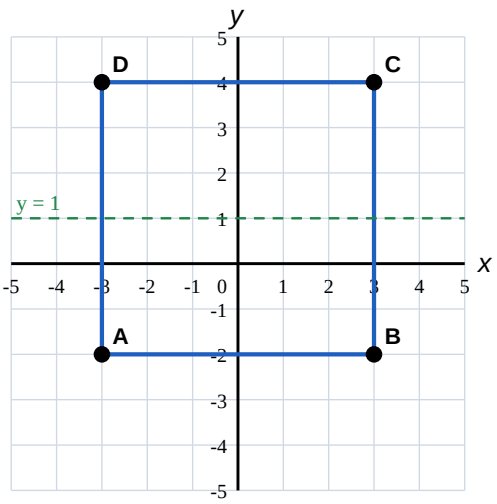
Key ideas: For $y = mx + c$, the gradient is m and the intercept is c. Lines with equal gradients are parallel. If $m > 0$ the line makes an acute angle with the positive x-axis; if $m < 0$ the angle is obtuse.

Question 1

(i), (ii) Side AB = 6 units, so C = **(3, 4)** and D = **(–3, 4)**.

(iii) AB: **y = –2** BC: **x = 3** CD: **y = 4** DA: **x = –3**

(iv) Axes of symmetry: **x = 0** (the y-axis) and **y = 1**.



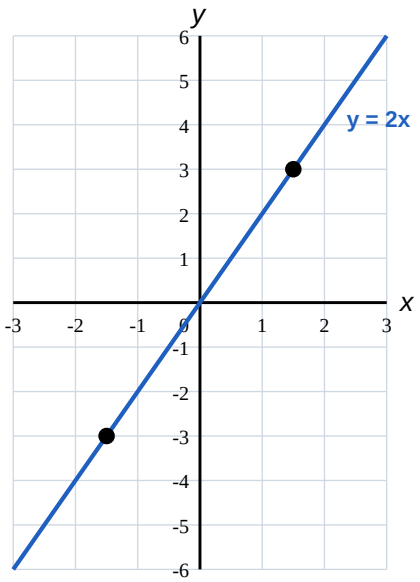
Question 2

x	-2	-1	0	1	2
y	-4	-2	0	2	4

(iii) When x = 1.5, **y = 3**

(iv) When y = –3, **x = –1.5**

(v) **No.** When x = –1, y = 2 × (–1) = –2, not 2.



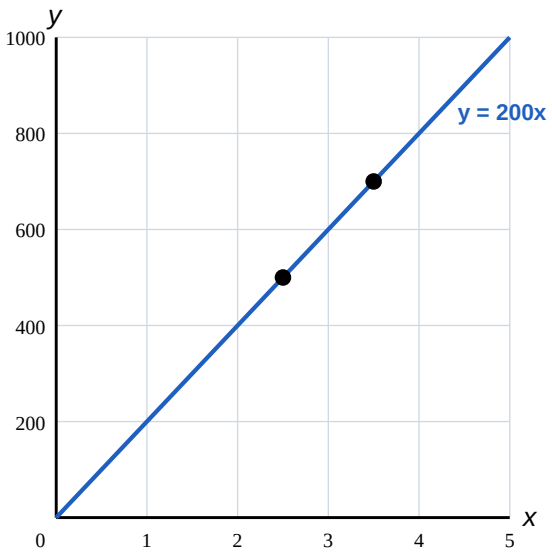
Question 3

(i) **y = 200x**

x (kg)	1	2	3	4
y (Rs)	200	400	600	800

(iv) Price of 2.5 kg = **Rs 500**

(v) Rs 700 buys **3.5 kg**



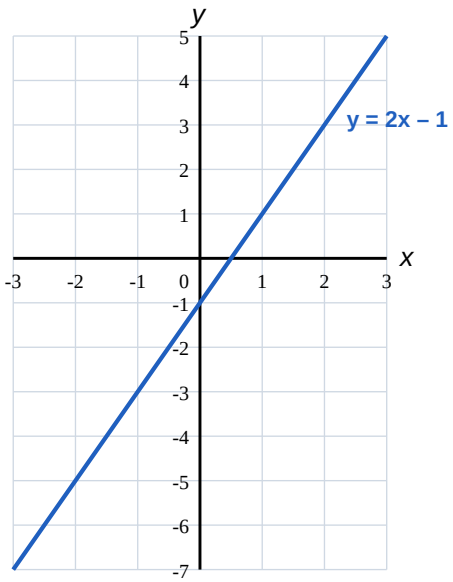
Question 4

x	-2	-1	0	1	2
y	-5	-3	-1	1	3

(iii) Gradient m = **2**, intercept c = **–1**

(iv) k = 2(4) – 1 = **7**

(v) Same gradient 2, intercept 2: **y = 2x + 2**



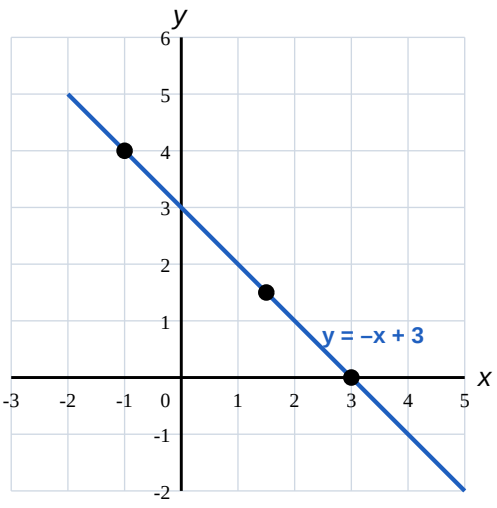
Question 5

x	-2	-1	0	1	2	3	4
y	5	4	3	2	1	0	-1

(iii) Intercept = **3**; meets the x-axis at **(3, 0)**

(iv) When x = 1.5, **y = 1.5**; when y = 4, **x = –1**

(v) **Obtuse**, because the gradient (–1) is negative.



Question 6

(i) (d) 2y = 4x + 6 → **y = 2x + 3** (e) y + 3x = 1 → **y = –3x + 1**

Line	(a)	(b)	(c)	(d)	(e)	(f)
Gradient m	3	–2	½	2	–3	3

Line	(a)	(b)	(c)	(d)	(e)	(f)
Intercept c	2	5	–4	3	1	–1

(iii) Acute (m positive): **(a), (c), (d), (f)** Obtuse (m negative): **(b), (e)**

(iv) **(a) and (f)** are parallel, because both have the same gradient, 3.

Question 7

(i) **y = 3x – 2**

(ii) **y = –x + 4**

(iii) Gradient 2 (parallel), c = –3 → **y = 2x – 3**

(iv) c = 0; m = 6 ÷ 2 = 3 → **y = 3x**

(v) c = 1; m = (5 – 1) ÷ (2 – 0) = 2 → **y = 2x + 1**

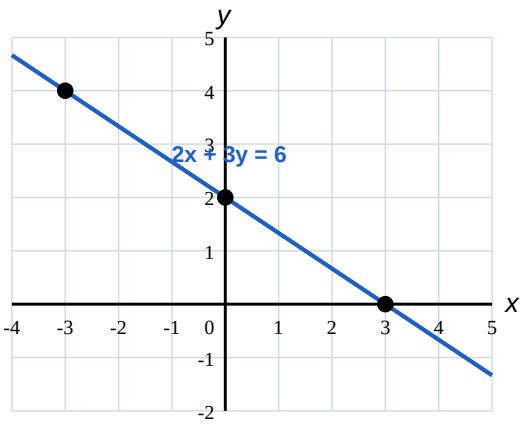
Question 8

(i) $3y = -2x + 6 \rightarrow$ **$y = -(2/3)x + 2$**

(ii) Gradient = **$-2/3$**, intercept = **2**

(iii) When $x = 0$, $y = 2 \rightarrow$ **(0, 2)**. When $y = 0$, $2x = 6$, $x = 3 \rightarrow$ **(3, 0)**

(v) $2(-3) + 3(4) = -6 + 12 = 6$, which equals the right-hand side, so (–3, 4) lies on the line.



Question 9

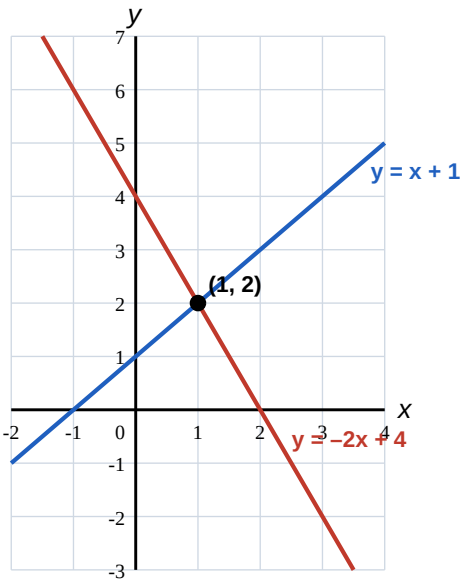
$y = x + 1$

x	-1	0	1	2	3
y	0	1	2	3	4

$y = -2x + 4$

x	-1	0	1	2	3
y	6	4	2	0	-2

(iii) $y = x + 1$: $m =$ **1**, $c =$ **1**. $y = -2x + 4$: $m =$ **-2**, $c =$ **4**

(iv) Point of intersection = **(1, 2)**

(v) **$x = 1$**